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1 Tensor, Symmetric, and Exterior Algebras

Let $\mathcal{T}^k(M) = \bigotimes_R^k M$, where $\mathcal{T}^0(M) = R$, $\mathcal{T}(M) = \bigoplus_{k < \omega} \mathcal{T}^k(M)$.

Suppose $\varphi : M \rightarrow A$ is an R -module homomorphism. Then there exists a unique $\Phi : \mathcal{T}(M) \rightarrow A$ such that $\Phi|M = \varphi$.

Note that $\Phi\left(\bigotimes_{i=1}^k m_i\right) = \prod_{i=1}^k \Phi(m_i) = \prod_{i=1}^k \varphi(m_i)$, so $\Phi|\mathcal{T}^k(M)$ is uniquely determined. Further, $\Phi(r) = r\Phi(1) = r$, so $\Phi|\mathcal{T}^0(M)$ is determined. Since $\mathcal{T}(M) = \bigoplus_{k < \omega} \mathcal{T}^k(M)$, $\Phi|\mathcal{T}(M)$ is uniquely determined, and we're done.

Suppose that $\mathcal{B}_1, \mathcal{B}_2$ are bases of V_1, V_2 . Then $\{v_1 \otimes v_2 : (v_1, v_2) \in \mathcal{B}_1 \times \mathcal{B}_2\}$ is a basis of $V_1 \otimes V_2$.

Ex: Take $R = \mathbb{Z}$ and $M = \mathbb{Q}/\mathbb{Z}$. Then $\mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} = 0$, so $\mathcal{T}(\mathbb{Q}/\mathbb{Z}) = \mathbb{Z} \oplus \mathbb{Q}/\mathbb{Z}$, so $(a, \bar{p}) \cdot (b, \bar{q}) = (ab, \overline{pq})$ and $(a, \bar{p}) + (b, \bar{q}) = (a+b, \overline{p+q})$.

Ex 2: Claim: $\mathbb{Z}/n\mathbb{Z} \otimes \mathbb{Z}/m\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z}$, where $d = \gcd(m, n)$. $\mathbb{Z}/n\mathbb{Z}$ has basis $\{\bar{1}\}$ and $\mathbb{Z}/m\mathbb{Z}$ also has basis $\{\bar{1}\}$. Then $\mathbb{Z}/m\mathbb{Z} \otimes \mathbb{Z}/n\mathbb{Z}$ has basis $\{\bar{1} \otimes \bar{1}\}$, which has order $d = \gcd(m, n)$.

Then $\mathcal{T}(\mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z} \oplus \bigoplus_{k < \omega} \mathbb{Z}/n\mathbb{Z}$.

Let R be commutative, A, B, M are R -modules. Then $\text{Hom}(A \times B, M) \cong \text{Hom}(A, M) \times \text{Hom}(B, M)$.

Proof: Let $i_A : A \rightarrow A \times B$ be given by $i_A(a) = (a, 0)$. Define i_B similarly. Define $\Phi : \text{Hom}(A \times B, M) \rightarrow \text{Hom}(A, M) \times \text{Hom}(B, M)$ by $\Phi(f) = (fi_A, fi_B)$. Suppose $f_A \in \text{Hom}(A, M)$ and $f_B \in \text{Hom}(B, M)$. Then $\Phi(f_A\pi_A + f_B\pi_B) = (f_A, f_B)$, so Φ is surjective.

Suppose $\Phi(f) = 0$. Then $fi_A = 0$ and $fi_B = 0$. Note $i_A\pi_A + i_B\pi_B = \text{id}_{A \times B}$, so $f = f\text{id}_{A \times B} = 0$.

Linearity should hold.

$$\text{Hom}(M, A \times B) \cong \text{Hom}(M, A) \times \text{Hom}(M, B).$$

Let $R = \mathbb{Z}$, $I = \mathbb{Z}^+$, and $M_i = \mathbb{Z}_i$. Then $\prod_{i \in I} M_i \not\cong \bigoplus_{i \in I} M_i$. Suppose $x \in \bigoplus_{i \in I} M_i$. There exists $N \in \mathbb{Z}^+$ such that for all $i > N$, we have $x_i = 0$. Pick $a \in \mathbb{Z}^+$ such that $ax = 0$. Then $x \in \text{Tor}(\bigoplus_{i \in I} M_i)$, so $\text{Tor}(\bigoplus_{i \in I} M_i) = \bigoplus_{i \in I} M_i$.

Consider $x = (1)_{i \in I}$. Suppose FTSOC that $x \in \text{Tor}(\prod_{i \in I} M_i)$, so there exists $a \in \mathbb{Z}^+$ such that $ax = 0$. Then $ax_{a+1} = \bar{a}$ in mod $a + 1$, absurd.

An arbitrary free product of free modules need not be free. Let $R = \mathbb{Z}$ and $M_i = \mathbb{Z}$. Let $M = \prod_{i \in I} M_i$ and $N = \bigoplus_{i \in I} M_i \leq M$. Suppose FTSOC that M has a basis $\mathcal{B}$.

Note that N is countable, so enumerate $N = \{x_1, x_2, x_3, \dots\}$. Since $M = \langle \mathcal{B} \rangle$, for each $x_i \in N$, there exists a finite subset $\mathcal{B}_i \subset \mathcal{B}$ such that $x_i \in \langle \mathcal{B}_i \rangle$. Let $\mathcal{B}' = \bigcup_{i \in I} \mathcal{B}_i$, which is countable. Let $N' = \langle \mathcal{B}' \rangle$, which is countable, free on $\mathcal{B}'$, and contains N .

Let $\overline{M} = M/N'$. I claim that $\overline{M}$ is a free $\mathbb{Z}$ -module. Sps FTSOC there exist nonzero a_i such that $\sum_{i=1}^n a_i \overline{y_i} = 0$ for $y_i \in M$, ie $\sum_{i=1}^n a_i y_i \in N'$. Since N' is free, we have $a_i = 0$ for all $i \in [n]$, absurd.

Suppose $\overline{x} \in \overline{M}$ is nonzero. Then I claim there exist finitely many $k \in \mathbb{Z}$ such that $\overline{x} = k\overline{m}$ for some $m \in M$. Write $\overline{x} = \sum a_i \overline{z_i}$ where the $\overline{z_i}$ are basis elements and the $a_i \neq 0$. Suppose $\overline{x} = k\overline{m}$. Write $\overline{m} = \sum b_i \overline{z_i}$ and expand the previous sum if necessary. Then we wish for

$$\sum (a_i - kb_i) \overline{z_i} = 0,$$

so $kb_i = a_i$ for all i . In particular, $k \mid a_i$ for some i , so there are finitely many such k (since $a_i \neq 0$) (and there are finitely many nonzero a_i).

Let $\mathcal{S} = \{(b_i)_{i \in I}\}$, where $b_i = \pm i!$. Note that $|\mathcal{S}| = 2^\omega$ is uncountable. Since N' is countable, there exists $x \in \mathcal{S}$ such that $x \notin N'$. Note that $\overline{x} \neq 0$ since each element of $\mathcal{S}$ is infinite. By the previous results, there exist finitely many k for which there exists $m \in M$ such that $x = k\overline{m}$.

However, observe that $\overline{x} = \overline{x'_n}$, where x'_n denotes the truncated sequence with $x'_{ni} = 0$ for $i < n$ and $x'_{ni} = x_{ni}$ for $i \geq n$. In particular, there exists $m_n \in M$ such that $x'_n = nm_n$, so there are infinitely many k , absurd.

2 Tensor Product of Modules

Suppose M is a left R -module and $N \leq M$ is a submodule. Then M/N is a left R -module with action given by $r\overline{m} = \overline{rm}$.

Proof: Suppose $m_1 - m_2 \in N$. Then $r(m_1 - m_2) \in N$, so $rm_1 - rm_2 \in N$. Hence, $R : M/N$ is well-defined. Further,

$$\begin{aligned} (r_1 + r_2)\overline{m} &= \overline{(r_1 + r_2)m} = \overline{r_1m + r_2m} = \overline{r_1m} + \overline{r_2m} = r_1\overline{m} + r_2\overline{m} \\ r(\overline{m_1} + \overline{m_2}) &= \overline{r(m_1 + m_2)} = \overline{rm_1 + rm_2} = \overline{rm_1} + \overline{rm_2} = r\overline{m_1} + r\overline{m_2} \\ (r_1r_2)\overline{m} &= \overline{(r_1r_2)m} = \overline{r_1(r_2m)} = r_1\overline{r_2m} = r_1(r_2\overline{m}), \end{aligned}$$

and if R has identity 1,

$$1\overline{m} = \overline{1m} = \overline{m},$$

so M/N indeed has a left R -module structure.

Let $R : M$, $R \leq S$ and let F be the free $\mathbb{Z}$ -module on $S \times M$. Let $S : F$ by $s_1 \cdot (s_2, r) = (s_1s_2, r)$. Then $S : S \otimes_R M$ by the results earlier.

Suppose $R \leq S$, $R : N$, $S : L$, and $\varphi : N \rightarrow L$ is a homomorphism of R -modules. Let $\iota : N \rightarrow S \otimes_R N$ be given by $\iota(n) = 1 \otimes n$. Then φ extends uniquely to $\Phi : S \otimes_R N \rightarrow L$ such that $\Phi\iota = \varphi$.

Proof: Let F denote the free module on $S \times N$. Let $\Phi_0 : F \rightarrow L$ be defined uniquely by $\Phi_0((s, n)) = s\varphi(n)$. Note that

$$\begin{aligned} F((s_1 + s_2, n) - (s_1, n) - (s_2, n)) &= (s_1 + s_2)\varphi(n) - s_1\varphi(n) - s_2\varphi(n) = 0 \\ F((s, n_1 + n_2) - (s, n_1) - (s, n_2)) &= s\varphi(n_1 + n_2) - s\varphi(n_1) - s\varphi(n_2) = 0 \\ F((sr, n) - (s, rn)) &= sr\varphi(n) - s\varphi(rn) = 0. \end{aligned}$$

Hence, Φ_0 descends to the homomorphism of R modules $\Phi : S \otimes_R N \rightarrow L$ such that $\Phi p = \Phi_0$, where $p : F \rightarrow S \otimes_R N$ is the projection.

Note that

$$s'\Phi(s \otimes n) = s'(s\varphi(n)) = (s's)\varphi(n) = \Phi(ss' \otimes n),$$

so Φ is actually a homomorphism of S -modules.

In particular,

$$\varphi(n) = \Phi_0((1, n)) = \Phi(1 \otimes n) = \Phi\iota(n),$$

so $\Phi\iota = \varphi$, showing existence.

Conversely, suppose Φ_1, Φ_2 satisfy the commutative diagram. Then $\Phi_1\iota = \Phi_2\iota = \varphi$, so

$$\Phi_1(s \otimes n) = \Phi_1(s(1 \otimes n)) = s\Phi_1\iota(n) = s\Phi_2\iota(n) = \Phi_2(s \otimes n)$$

Since the elementary factors $s \otimes n$ generate $S \otimes_R N$, we have that $\Phi_1 = \Phi_2$, showing uniqueness.

If $\Phi : S \otimes_R N \rightarrow L$ is any homomorphism of R -modules, then so is $\Phi\iota : N \rightarrow L$.

Let $\mathcal{M}$ be a collection of R -modules. Write $M \preceq N$ if there exists $N' \leq N$ such that $M \cong N/N'$. I claim that $(\mathcal{M}, \preceq)$ is a partial order.

Note that $M \cong M/0$, so $M \preceq M$.

Let M, N be right, left R -modules and L an abelian group. Let $\varphi : M \times N \rightarrow L$ be R -balanced. Suppose $f : L \rightarrow L'$ is a group homomorphism. Then $f\varphi$ is R -balanced.

Proof:

$$\begin{aligned} f\varphi(m_1 + m_2, n) &= f(\varphi(m_1, n) + \varphi(m_2, n)) = f\varphi(m_1, n) + f\varphi(m_2, n) \\ f\varphi(m, n_1 + n_2) &= f(\varphi(m, n_1) + \varphi(m, n_2)) = f\varphi(m, n_1) + f\varphi(m, n_2) \\ f\varphi(mr, n) &= f\varphi(m, rn) \end{aligned}$$

Hence, $f\varphi$ is R -balanced.

Let $\iota : M \times N \rightarrow M \otimes_R N$ be given by $\iota(m, n) = m \otimes n$. I claim that ι is R -balanced.

$$\begin{aligned} \iota(m_1 + m_2, n) &= (m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n = \iota(m_1, n) + \iota(m_2, n) \\ \iota(m, n_1 + n_2) &= \iota(m, n_1) + \iota(m, n_2) \\ \iota(mr, n) &= mr \otimes n = m \otimes rn = \iota(m, rn). \end{aligned}$$

It follows that $\Phi\iota$ is R -balanced.

Conversely, suppose that $\varphi : M \times N \rightarrow L$ is R -balanced. Extend φ to $\psi : F \rightarrow L$, where F is the free $\mathbb{Z}$ module on $M \times N$, by $\psi((m, n)) = \varphi(m, n)$. Note that

$$\begin{aligned} \psi((m_1 + m_2, n) - (m_1, n) - (m_2, n)) &= \varphi(m_1 + m_2, n) - \varphi(m_1, n) - \varphi(m_2, n) = 0 \\ \psi((m, n_1 + n_2) - (m, n_1) - (m, n_2)) &= \varphi(m, n_1 + n_2) - \varphi(m, n_1) - \varphi(m, n_2) = 0 \\ \psi((mr, n) - (rm, n)) &= \varphi(mr, n) - \varphi(rm, n) = 0. \end{aligned}$$

Hence, ψ descends to the map $\Phi : M \otimes_R N \rightarrow L$ such that $\varphi = \Phi\iota$.

$$(ss') \cdot r = (ss')f(r) = s(s'f(r)) = s(s' \cdot r).$$

Suppose $S : M$ and $R \leq Z(S)$. Then $M : R$ and

$$(sm)r = r(sm) = (rs)m = (sr)m = s(rm) = s(mr).$$

Fix $s \in S$ and consider the map $M \times N \rightarrow M \oplus_R N$ by $(m, n) \mapsto sm \oplus n$. We have $s(m+m') \oplus n = (sm+sm') \oplus n = sm \oplus n + sm' \oplus n$. $sm \oplus (n+n') = sm \oplus n + sm \oplus n'$. $(sm)r \oplus n = sm \oplus rn$. Let $\lambda_s : M \oplus_R N \rightarrow M \oplus_R N$ such that $\lambda_s(m \oplus n) = sm \oplus n$.

Suppose $\varphi : M \times N \rightarrow L$ is R -bilinear. In particular,

$$\varphi(mr, n) = \varphi(rm, n) = r\varphi(m, n) = \varphi(m, rn),$$

so φ is R -balanced. Let $\Phi : M \otimes_R N \rightarrow L$ be the induced homomorphism of R -modules.

Now suppose $\Phi : M \otimes_R N \rightarrow L$ is a R -module homomorphism. Then

$$\begin{aligned} \Phi\iota(a_1m_1 + a_2m_2, n) &= a_1\Phi\iota(m_1, n) + a_2\Phi\iota(m_2, n) \\ \Phi\iota(m, b_1n_1 + b_2n_2) &= b_1\Phi\iota(m, n_1) + b_2\Phi\iota(m, n_2). \end{aligned}$$

Let M, M' be right R -modules, N, N' be left R -modules, $\varphi : M \rightarrow M'$, $\psi : N \rightarrow N'$ are module homomorphisms. Then there is a unique group homomorphism $\varphi \otimes \psi : M \otimes_R N \rightarrow M' \otimes_R N'$ such that $(\varphi \otimes \psi)(m \otimes n) = \varphi(m) \otimes \psi(n)$ for all $m \in M$ and $n \in N$.

Define $F : M \times N \rightarrow M' \otimes_R N'$ by $F(m, n) = \varphi(m) \otimes \psi(n)$. Note that

$$\begin{aligned} F(m_1 + m_2, n) &= \varphi(m_1 + m_2) \otimes \psi(n) = (\varphi(m_1) + \varphi(m_2)) \otimes \psi(n) \\ &= \varphi(m_1) \otimes \psi(n) + \varphi(m_2) \otimes \psi(n) = F(m_1, n) + F(m_2, n) \\ F(m, n_1 + n_2) &= F(m, n_1) + F(m, n_2) \\ F(mr, n) &= \varphi(mr) \otimes \psi(n) = \varphi(m)r \otimes \psi(n) = \varphi(m) \otimes r\psi(n) \\ &= \varphi(m) \otimes \psi(rn) = F(m, rn), \end{aligned}$$

so F is R -balanced. Let $\varphi \otimes \psi : M \otimes_R N \rightarrow M' \otimes_R N'$ be such that $F = (\varphi \otimes \psi)\iota$, where $\iota : M \times N \rightarrow M \otimes_R N$ is the R -balanced map given by $\iota(m, n) = m \otimes n$.

In particular, $\varphi(m) \otimes \psi(n) = F(m, n) = (\varphi \otimes \psi)(m \otimes n)$ for all $m \in M, n \in N$. Note that the simple products $m \otimes n$ span $M \otimes N$, so $\varphi \otimes \psi$ is determined.

Sps further that M, M' are (S, R) bimodules. Then $\varphi \otimes \psi$ is a homomorphism of left S -modules.

Proof: We have

$$(\varphi \otimes \psi)(sm \otimes n) = \varphi(sm) \otimes \psi(n) = s(\varphi(m) \otimes \psi(n)) = s((\varphi \otimes \psi)(m \otimes n)).$$

Extending by linearity to arbitrary elements of $M \otimes N$, we have that $\varphi \otimes \psi$ is a S -module homomorphism.

Sps $\lambda : M' \rightarrow M'', \mu : N' \rightarrow N''$ are R -module homomorphisms. Then $(\lambda \otimes \mu) \circ (\varphi \otimes \psi) = (\lambda \circ \varphi) \otimes (\mu \circ \psi)$.

Proof:

$$\begin{aligned} (\lambda \otimes \mu) \circ (\varphi \otimes \psi)(m, n) &= (\lambda \otimes \mu)(\varphi(m) \otimes \psi(n)) \\ &= \lambda\varphi(m) \otimes \mu\psi(n) = (\lambda \circ \varphi) \otimes (\mu \circ \psi)(m, n). \end{aligned}$$

Sps M is right R -module, N is (R, T) bimodule, L is left T -module. Then there is a unique group isomorphism $(M \otimes_R N) \otimes_T L \rightarrow M \otimes_R (N \otimes_T L)$ such that $(m \otimes n) \otimes l \mapsto m \otimes (n \otimes l)$.

Fix $l \in L$. Define $F_l : M \times N \rightarrow M \otimes_R (N \otimes_T L)$ by $F_l(m, n) = m \otimes (n \otimes l)$. We have

$$\begin{aligned} F_l(m_1 + m_2, n) &= (m_1 + m_2) \otimes (n \otimes l) = m_1 \otimes (n \otimes l) + m_2 \otimes (n \otimes l) \\ &= F_l(m_1, n) + F_l(m_2, n) \\ F_l(m, n_1 + n_2) &= m \otimes ((n_1 + n_2) \otimes l) = m \otimes (n_1 \otimes l + n_2 \otimes l) \\ &= m \otimes (n_1 \otimes l) + m \otimes (n_2 \otimes l) = F_l(m, n_1) + F_l(m, n_2). \\ F_l(mr, n) &= mr \otimes (n \otimes l) = m \otimes (rn \otimes l) = F_l(m, rn), \end{aligned}$$

so F_l is R -balanced. Let the unique induced group homomorphism be $\Phi_l : M \otimes_R N \rightarrow M \otimes_R (N \otimes_T L)$.

Define $F : (M \otimes_R N) \times L \rightarrow M \otimes_R (N \otimes_T L)$ by $F(x, l) = \Phi_l(x)$ for all $x \in M \otimes_R N$ and $l \in L$.

We have

$$F(x_1 + x_2, l) = \Phi_l(x_1 + x_2) = \Phi_l(x_1) + \Phi_l(x_2) = F(x_1, l) + F(x_2, l).$$

Note that

$$\begin{aligned} F(m \otimes n, l_1 + l_2) &= \Phi_{l_1 + l_2}(m \otimes n) = F_{l_1 + l_2}(m, n) \\ &= m \otimes (n \otimes (l_1 + l_2)) = m \otimes (n \otimes l_1 + n \otimes l_2) \\ &= m \otimes (n \otimes l_1) + m \otimes (n \otimes l_2) = F(m \otimes n, l_1) + F(m \otimes n, l_2), \end{aligned}$$

so

$$\begin{aligned}
F\left(\sum m_i \otimes n_i, l_1 + l_2\right) &= \sum F(m_i \otimes n_i, l_1 + l_2) \\
&= \sum F(m_i \otimes n_i, l_1) + F(m_i \otimes n_i, l_2) \\
&= F\left(\sum m_i \otimes n_i, l_1\right) + F\left(\sum m_i \otimes n_i, l_2\right).
\end{aligned}$$

Finally,

$$\begin{aligned}
F((m \otimes n)t, l) &= \Phi_l(m \otimes nt) = m \otimes (nt \otimes l) \\
&= m \otimes (n \otimes tl) = F(m \otimes n, tl),
\end{aligned}$$

so F is T -balanced. Hence, F descends to the unique map $\Phi : (M \otimes_R N) \otimes_T L \rightarrow M \otimes_R (N \otimes_T L)$ such that $\Phi((m \otimes n) \otimes l) = \Phi_l(m \otimes n) = m \otimes (n \otimes l)$ for all $(m, n, l) \in M \times N \times L$.

I claim that Φ is bijective. To show this, construct the reverse map $M \otimes_R (N \otimes_T L) \rightarrow (M \otimes_R N) \otimes_T L$ such that $m \otimes (n \otimes l) \mapsto (m \otimes n) \otimes l$. Then the maps are inverses, so Φ is a group isomorphism.

Sps that M is a (S, R) bimodule. I claim that $M \otimes_R N$ is a (S, T) bimodule.

Proof: $(s(m \otimes n))t = (sm \otimes n)t = sm \otimes nt$ and $s((m \otimes n)t) = s(m \otimes nt) = sm \otimes nt$. Extend from simple products $m \otimes n$ to $M \otimes N$. Then $(M \otimes_R N) \otimes_T L$ is a left S -module.

$N \otimes_T L$ is a left R -module. Then $M \otimes_R (N \otimes_T L)$ is a left S -module.

If R is commutative, M, N, L are left R -modules, then $(M \otimes N) \otimes L \cong M \otimes (N \otimes L)$ by the usual module structure.

Sps $\varphi : M \rightarrow L$ be a homomorphism of left R -modules. Sps $M = M_1 \oplus M_2$. Let $\varphi_1 = \varphi|_{M_1}, \varphi_2 = \varphi|_{M_2}$. There is a bijection $\varphi \mapsto (\varphi_1, \varphi_2)$.

$n = 1$ holds. Sps the result for n . Let $\varphi_{m_{n+1}} : \prod_{i=1}^n M_i \rightarrow L$ be φ with the $n + 1$ th argument fixed as m_{n+1} . By the IH, there is a unique R -module homomorphism $\Phi_{m_{n+1}} : \bigotimes_{i=1}^n M_i \rightarrow L$ such that $\Phi_{m_{n+1}} i_n = \varphi_{m_{n+1}}$.

For $(x_n, m_{n+1}) \in \prod_{i=1}^{n+1} M_i$, define $F : \bigotimes_{i=1}^n M_i \times M_{n+1} \rightarrow L$ by $F(x_n, m_{n+1}) =$

$\Phi_{m_{n+1}}(x_n)$. Note that

$$\begin{aligned}
F(x_n + x'_n, m_{n+1}) &= \Phi_{m_{n+1}}(x_n + x'_n) = \Phi_{m_{n+1}}(x_n) + \Phi_{m_{n+1}}(x'_n) \\
&= F(x_n, m_{n+1}) + F(x'_n, m_{n+1}) \\
F(\otimes_{i=1}^n m_i, m_{n+1} + m'_{n+1}) &= \Phi_{m_{n+1} + m'_{n+1}}(\otimes_{i=1}^n m_i) = \Phi_{m_{n+1} + m'_{n+1}} \iota_n((m_i)_{i \in [n]}) \\
&= \varphi((m_i)_{i \in [n]} \times (m_{n+1} + m'_{n+1})) = \varphi((m_i)_{i \in [n]} \times m_{n+1}) + \varphi((m_i)_{i \in [n]} \times m'_{n+1}) \\
&= F(\otimes_{i=1}^n m_i, m_{n+1}) + F(\otimes_{i=1}^n m_i, m'_{n+1}) \\
F(rx_n, m_{n+1}) &= \Phi_{m_{n+1}}(rx_n) = r\Phi_{m_{n+1}}(x_n) = rF(x_n, m_{n+1}) \\
F(\otimes_{i=1}^n m_i, rm_{n+1}) &= \varphi((m_i)_{i \in [n]} \times rm_{n+1}) = r\varphi((m_i)_{i \in [n]} \times m_{n+1}) \\
&= rF(\otimes_{i=1}^n m_i, m_{n+1}).
\end{aligned}$$

Hence, there is a unique homomorphism $\Phi : \bigotimes_{i=1}^{n+1} M_i \rightarrow L$ such that $\Phi \iota' = F$, where $\iota' : \bigotimes_{i=1}^n M_i \times M_{n+1} \rightarrow \bigotimes_{i=1}^{n+1} M_i$.

Apply $\Phi \iota'(\iota_n \times \text{id}_{M_{n+1}}) = F(\iota_n \times \text{id}_{M_{n+1}})$, or $\Phi \iota_{n+1} = \varphi$, as desired.

Let M, M' be right R -modules, N, N' be left R -modules. Then there are unique group isomorphisms

$$\begin{aligned}
(M \oplus M') \otimes_R N &\cong (M \otimes_R N) \oplus (M' \otimes_R N) \\
M \otimes_R (N \oplus N') &\cong (M \otimes_R N) \oplus (M \otimes_R N')
\end{aligned}$$

such that $(m, m') \otimes n \mapsto (m \otimes n, m' \otimes n)$ and $m \otimes (n, n') \mapsto (m \otimes n, m \otimes n')$.

Let $((m, m'), n) \mapsto (m \otimes n, m' \otimes n)$. We have

$$\begin{aligned}
&((m_1 + m_2, m'_1 + m'_2), n) - ((m_1, m'_1), n) - ((m_2, m'_2), n) \mapsto \\
&((m_1 + m_2) \otimes n, (m'_1 + m'_2) \otimes n) - (m_1 \otimes n, m'_1 \otimes n) - (m_2 \otimes n, m'_2 \otimes n) = 0 \\
&((m_1, m_2)r, n) - ((m_1, m_2), rn) \mapsto (m_1 r \otimes n, m_2 r \otimes n) - (m_1 \otimes rn, m_2 \otimes rn) = 0,
\end{aligned}$$

so this map is R -balanced. Let the unique induced map be $(M \oplus M') \otimes N \rightarrow (M \otimes N) \oplus (M' \otimes N)$.

$\text{Sps}((m, m'), n) \mapsto 0$. Then $m \otimes n = 0$ and $m' \otimes n = 0$. Write

$$((m, m'), n) = [((m, m'), n) - ((m, 0), n) - ((0, m'), n)] + ((m, 0), n) + ((0, m'), n).$$

We have $(m, n) = \sum(m_1 + m_2, n) - (m_1, n) - (m_2, n) + \sum(m, n_1 + n_2) - (m, n_1) - (m, n_2) + \sum(mr, n) - (m, rn)$. Note that $((m, 0), n) = \sum((m_1 + m_2, 0), n) - ((m_1, 0), n) - ((m_2, 0), n) + \sum((m, 0), n_1 + n_2) - ((m, 0), n_1) - ((m, 0), n_2) + \sum((mr, 0r), n) - \sum((m, 0), rn)$. Hence, $(m, m') \otimes n = 0$.

Note that $((m, 0), n) \mapsto (m \otimes n, 0 \otimes n) = (m \otimes n, 0)$, so $M \otimes N \hookrightarrow (M \otimes N) \oplus (M' \otimes N)$ is in the image. Similarly, $M' \otimes N$ is in the image, so the map is surjective by additivity.

Sps further that M, M' are left (S, R) bimodules. Then $(s(m, m'), n) \mapsto (sm \otimes n, sm' \otimes n) = s(m \otimes n, m' \otimes n)$, so the map is a left S -module isomorphism.

Let $R \leq S$. I claim that $S \otimes_R R \cong S$.

Proof: Define $(s, r) \mapsto sr$. We have

$$\begin{aligned} (s_1 + s_2, r) - (s_1, r) - (s_2, r) &\mapsto 0 \\ (s, r_1 + r_2) - (s, r_1) - (s, r_2) &\mapsto 0 \\ (sr', r) - (s, r'r) &\mapsto (sr')r - s(r'r) = 0, \end{aligned}$$

so we have the map $S \otimes_R R \mapsto S$. Note that $s \otimes 1 \mapsto s$, so the map is surjective. Suppose $s \otimes r \mapsto 0$. Then $sr = 0$, so $s \otimes r = sr \otimes 1 = 0 \otimes 1 = 0$, so the map is injective.

Note that $s'(s, r) = (ss', r) \mapsto (ss')r = s(s'r)$, so the map is an isomorphism of left S -modules.

Corr: $S \otimes_R R^n \cong (S \otimes_R R)^n \cong S^n$.

Corr: Let R a commutative ring, $M \cong R^s$, $N \cong R^t$ with bases $\{e_i\}_{i \in [s]}$ and $\{f_j\}_{j \in [t]}$. Then $M \otimes N \cong R^{st}$ with basis $\{e_i \otimes f_j\}_{(i,j) \in [s] \times [t]}$.

$$R^s \otimes_R R^t \cong (R \otimes_R R^t)^s \cong (R^t)^s \cong R^{st}.$$

Let R be commutative, M, N with R -module structures. Then there exists a unique R -module isomorphism $M \otimes_R N \rightarrow N \otimes_R M$ given by $m \otimes n \mapsto n \otimes m$.

Proof: Define the map $M \times N \rightarrow M \otimes N$ by $(m, n) \mapsto n \otimes m$. This map is R -bilinear, so we have the unique R -module homomorphism $\Phi : M \otimes_R N \rightarrow N \otimes_R M$ given by $\Phi(m \otimes n) = n \otimes m$.

It is easy to define the reverse map.

Let R be commutative, A, B be R -algebras. Then multiplication $(a \otimes b)(a' \otimes b') = aa' \otimes bb'$ is well-defined and maps $A \otimes B$ into an R -algebra.

Let $f_A : R \rightarrow A$ be such that $f_A(1_R) = 1_A$ and $f_A(R) \leq Z(A)$. Let f_B be defined similarly.

Define $\varphi : A \times B \times A \times B \rightarrow A \otimes_R B$ by $p(a, b, a', b') = aa' \otimes bb'$. Note

that p is additive with respect to any argument. Further,

$$\begin{aligned}\varphi(ra, b, a', b') &= raa' \otimes bb' = r\varphi(a, b, a', b') \\ \varphi(a, rb, a', b') &= aa' \otimes rbb' = r\varphi(a, b, a', b') \\ \varphi(a, b, ra', b') &= ara' \otimes bb' = r\varphi(a, b, a', b') \\ \varphi(a, b, a', rb') &= aa' \otimes brb' = r\varphi(a, b, a', b'),\end{aligned}$$

so p is 4-linear.

Let the unique induced homomorphism of R -modules $\Phi : A \otimes_R B \otimes_R A \otimes_R B \rightarrow A \otimes_R B$ be given by $\Phi(a \otimes b \otimes a' \otimes b') = aa' \otimes bb'$. By associativity, identify $A \otimes_R B \otimes_R A \otimes_R B$ by $(A \otimes_R B) \otimes_R (A \otimes_R B)$. By the correspondence, we have an R -bilinear map $F : (A \otimes_R B) \times (A \otimes_R B) \rightarrow A \otimes_R B$ such that $F(a \otimes b, a' \otimes b') = aa' \otimes bb'$, which I claim is our desired multiplication operation on $A \otimes_R B$.

Indeed, additivity of F gives that multiplication is distributive. Define $f : R \rightarrow A \otimes_R B$ by $f(r) = r(1_A \otimes 1_B)$. Note that $f(1_R) = 1_A \otimes 1_B$ is the identity of $A \otimes_R B$ since $\Phi(1_A \otimes 1_B, a \otimes b) = \Phi(a \otimes b, 1_A \otimes 1_B) = a \otimes b$ (extend to $A \otimes_R B$ by linearity).

Further,

$$\Phi(r(1_A \otimes 1_B), a \otimes b) = r(a \otimes b) = \Phi(a \otimes b, r(1_A \otimes 1_B)),$$

so $f(R) \leq Z(A \otimes_R B)$.

Finally,

$$\begin{aligned}f(rs) &= rs(1_A \otimes 1_B) = rs \otimes 1_B = (r \otimes 1_A)(s \otimes 1_B) = f(r)f(s) \\ f(r+s) &= (r+s)(1_A \otimes 1_B) = f(r) + f(s),\end{aligned}$$

so f is a ring homomorphism.

Show that $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ and $\mathbb{C} \otimes_{\mathbb{C}} \mathbb{C}$ both left $\mathbb{R}$ -modules but not isomorphic as $\mathbb{R}$ -modules.

$\mathbb{C} \otimes_{\mathbb{C}} \mathbb{C} \cong \mathbb{C} \cong \mathbb{R}^2$. $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{R}^2 \otimes_{\mathbb{R}} \mathbb{R}^2 \cong \mathbb{R}^4$, so because the vector spaces have different dimensions, they are non-isomorphic.

Define $\varphi : \mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q} \rightarrow \mathbb{Q}$ by $\varphi(q, r) = qr$. φ is bilinear, so we have the induced left $\mathbb{Q}$ -module homomorphism $\Phi : \mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q} \rightarrow \mathbb{Q}$.

Define $\Psi : \mathbb{Q} \rightarrow \mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ by $\Psi(q) = q \otimes 1$. Note that $\Phi\Psi(q) = \Phi(q \otimes 1) = q$ and $\Psi\Phi(q \otimes r) = \Psi(qr) = qr \otimes 1 = q(r \otimes 1)$

Define $f : \mathbb{Q} \rightarrow \mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ by $f(q) = q \otimes 1$. Note that $f(qr) = qr \otimes 1 = q(r \otimes 1) = qf(r)$.

Let R be an integral domain, F the field of fractions of R . Then $(F/R) \otimes_R (F/R) = 0$.

Proof: We have

$$\begin{aligned} \overline{a/b} \otimes \overline{c/d} &= \overline{a/b} \otimes \overline{c/(bd)} \cdot \bar{b} = \overline{a/b} \cdot \bar{b} \otimes \overline{c/(bd)} \\ &= \bar{a} \cdot \overline{c/(bd)} = 0 \cdot \overline{c/(bd)} = 0. \end{aligned}$$

Let R an integral domain, F field of fractions of R , N left R -module. Show that any element of $F \otimes_R N$ is of the form $\frac{1}{d} \otimes n$ for $d \neq 0$ in R and $n \in N$.

Proof: Let $a/b \otimes n, c/d \otimes m \in F \otimes_R N$ be simple tensors. Then

$$\begin{aligned} a/b \otimes n + c/d \otimes m &= \frac{1}{bd} \otimes adn + \frac{1}{bd} \otimes cbm \\ &= \frac{1}{bd} \otimes (adn + cbm), \end{aligned}$$

as desired.

Let R be an integral domain, $U = R^\times$, N a left R -module, Q the quotient field of R . Let $U^{-1}N = \{(u, n) : u \in U, n \in N\}$ under the equivalence $\sim$ given by $(u, n) \sim (u', n')$ iff there exists $v \in U$ such that $vun' = vn'n$.

Note $(u, n) \sim (u, n)$ since $1un = 1un$. Note if $(u, n) \sim (u', n')$, then $vun' = vu'n$, so $vu'n = vun'$, so $(u', n') \sim (u, n)$, so $\sim$ is reflexive, symmetric.

Suppose $(u, n) \sim (u', n')$, $(u', n') \sim (u'', n'')$. Then $vun' = vu'n$ and $wu'n'' = wu''n'$, so $vwuu''n' = vwu'u''n \rightarrow vwuu'n'' = vwu'u''n \rightarrow (vwu')un'' = (vwu')u''n$, so $(u, n) \sim (u'', n'')$ (since $vwu' \neq 0$), so $\sim$ is transitive.

Define the map $U^{-1}N \rightarrow Q \otimes_R N$ by $\overline{(u, n)} \mapsto 1/u \otimes n$. Suppose $(u, n) \sim (u', n')$, so $wun' = wu'n$. Then

$$1/u \otimes n = \frac{1}{uwu'} \otimes wu'n = \frac{1}{uwu'} \otimes wun' = 1/u' \otimes n',$$

so the map is well-defined.

Note

$$\begin{aligned} a/b \otimes n &\mapsto \overline{(b, an)} = \overline{a(b, n)} \mapsto a(1/b \otimes n) = a/b \otimes n, \\ \overline{(u, n)} &\mapsto 1/u \otimes n \mapsto \overline{(u, 1n)} = \overline{(u, n)}, \end{aligned}$$

so the maps are inverses.

Now suppose $1/d \otimes n = 0$ in $Q \otimes_R N$. Then $\overline{(d, n)} = 0$ in $U^{-1}N$, so $(d, n) \sim (1, 0)$, so there exists $r \in U$ such that $rd0 = r1n$, ie $rn = 0$.

Suppose $\mathbb{Q} \otimes_{\mathbb{Z}} A = 0$ for an abelian group A . Let $a \in A$ be arbitrary. From $1 \otimes a = 0$, we have $na = 0$ for some $n \neq 0$, so $\text{Tor}A = A$.

Consider the $\mathbb{Z}$ -module $M = \prod_{i \in \mathbb{Z}^+} \mathbb{Z}/2^i \mathbb{Z}$. Consider $(1)_{i \in \mathbb{Z}^+}$, which has infinite order. Then $\mathbb{Q} \otimes_{\mathbb{Z}} M \neq 0$ since $\text{Tor}M \neq M$, so direct products do not commute with tensor product.